

BUILDCORE Construction Portal

Step-by-step production installation guide — Ubuntu 24.04 LTS + Nginx
Stack: FastAPI (Python) · PostgreSQL · React · Nginx reverse proxy

Architecture Overview

The portal has three parts deployed on a single Ubuntu 24.04 server:

- **React frontend** — compiled into static files, served directly by Nginx
- **FastAPI backend** — runs on 127.0.0.1:8001 as a systemd service, proxied by Nginx under `/api`
- **PostgreSQL 16** — local database (`construction_db`)

Request flow: Browser → Nginx (port 80/443) → static React files, and /api/ → uvicorn on port 8001 → PostgreSQL.*

Step 1 — Prepare the server

Log in as a sudo-capable user and update the system:

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y git curl build-essential ufw
```

Configure the firewall (allow SSH + web traffic only):

```
sudo ufw allow OpenSSH
sudo ufw allow 'Nginx Full'
sudo ufw enable
sudo ufw status
```

Step 2 — Install PostgreSQL and create the database

```
sudo apt install -y postgresql postgresql-contrib
sudo systemctl enable --now postgresql
```

Create the database user and database (change the password!):

```
sudo -u postgres psql <<'SQL'
CREATE USER buildcore WITH PASSWORD 'CHANGE_ME_STRONG_PASSWORD';
CREATE DATABASE construction_db OWNER buildcore;
GRANT ALL PRIVILEGES ON DATABASE construction_db TO buildcore;
SQL
```

Verify the connection:

```
psql "postgresql://buildcore:CHANGE_ME_STRONG_PASSWORD@localhost:5432/construction_db" -c "SELECT 1;"
```

Step 3 — Install Python 3.12 and deploy the backend

```
sudo apt install -y python3 python3-venv python3-pip
sudo mkdir -p /opt/buildcore && sudo chown $USER:$USER /opt/buildcore
cd /opt/buildcore
# copy or clone your project here so you have /opt/buildcore/backend and /opt/buildcore/frontend
git clone <YOUR_REPO_URL> .
```

Create the virtual environment and install dependencies:

```
cd /opt/buildcore/backend
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

Create the backend environment file **/opt/buildcore/backend/.env**:

```
DATABASE_URL="postgresql+psycopg2://buildcore:CHANGE_ME_STRONG_PASSWORD@localhost:5432/construction_db"
JWT_SECRET="GENERATE_A_RANDOM_64_CHAR_HEX_STRING"
ADMIN_EMAIL="kesari4416@gmail.com"
ADMIN_PASSWORD="CHANGE_ME_ADMIN_PASSWORD"
CORS_ORIGINS="https://your-domain.com"
```

Generate a strong JWT secret with: openssl rand -hex 32

On first start the app creates all tables automatically and seeds the admin account. Alternatively run the included Alembic migration: alembic upgrade head

Step 4 — Run the backend as a systemd service

Create **/etc/systemd/system/buildcore-backend.service**:

```
[Unit]
Description=BUILDCORE FastAPI backend
After=network.target postgresql.service
Requires=postgresql.service

[Service]
User=www-data
Group=www-data
WorkingDirectory=/opt/buildcore/backend
EnvironmentFile=/opt/buildcore/backend/.env
ExecStart=/opt/buildcore/backend/venv/bin/uvicorn server:app \
    --host 127.0.0.1 --port 8001 --workers 2
Restart=always
RestartSec=3

[Install]
WantedBy=multi-user.target
```

Set permissions, enable and start the service:

```
sudo chown -R www-data:www-data /opt/buildcore/backend/uploads
sudo systemctl daemon-reload
sudo systemctl enable --now buildcore-backend
sudo systemctl status buildcore-backend
curl -s http://127.0.0.1:8001/api/ # expect: {"message":"Construction Portal API"}
```

Step 5 — Build the React frontend

Install Node.js 20 LTS and Yarn:

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -  
sudo apt install -y nodejs  
sudo npm install -g yarn
```

Point the frontend to your domain in **/opt/buildcore/frontend/.env**:

```
REACT_APP_BACKEND_URL=https://your-domain.com
```

Install dependencies and create the production build:

```
cd /opt/buildcore/frontend  
yarn install  
yarn build  
# static files are generated in /opt/buildcore/frontend/build
```

Step 6 — Install and configure Nginx

```
sudo apt install -y nginx
```

Create the site config **/etc/nginx/sites-available/buildcore**:

```

server {
    listen 80;
    server_name your-domain.com;

    root /opt/buildcore/frontend/build;
    index index.html;

    client_max_body_size 15M;    # allow photo uploads

    # React SPA – always fall back to index.html
    location / {
        try_files $uri /index.html;
    }

    # Long cache for hashed static assets
    location /static/ {
        expires 30d;
        add_header Cache-Control "public, immutable";
    }

    # API + uploaded photos -> FastAPI on port 8001
    location /api/ {
        proxy_pass http://127.0.0.1:8001;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_read_timeout 60s;
    }
}

```

Enable the site and reload Nginx:

```

sudo ln -s /etc/nginx/sites-available/buildcore /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t
sudo systemctl reload nginx

```

Step 7 — Enable HTTPS with Let's Encrypt (recommended)

The app sets secure cookies (Secure; SameSite=None), so HTTPS is required in production.

```

sudo apt install -y certbot python3-certbot-nginx
sudo certbot --nginx -d your-domain.com
sudo certbot renew --dry-run    # verify auto-renewal

```

Step 8 — Verify the deployment

```
# 1. Backend health
curl -s https://your-domain.com/api/

# 2. Login as admin (returns user JSON + token)
curl -s -X POST https://your-domain.com/api/auth/login \
  -H "Content-Type: application/json" \
  -d '{"email":"kesari4416@gmail.com","password":"YOUR_ADMIN_PASSWORD"}'

# 3. Open https://your-domain.com in a browser and sign in
```

Troubleshooting

502 Bad Gateway — backend not running. Check: `sudo journalctl -u buildcore-backend -n 50`

Database connection refused — check PostgreSQL: `sudo systemctl status postgresql`, and verify `DATABASE_URL` credentials in `.env`

Login works but session drops — you are serving over plain HTTP; secure cookies require HTTPS (complete Step 7)

Photo upload fails (413) — increase `client_max_body_size` in the Nginx config and reload

Blank page after deploy — rebuild the frontend after changing `.env`: `yarn build`, then hard-refresh. `REACT_APP_BACKEND_URL` is baked in at build time.

Nginx logs — `/var/log/nginx/access.log` and `/var/log/nginx/error.log`